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UNIZIK FRESHER EDITION | 2026

CHM 101: General Chemistry I - Atoms, Bonding and Reactions

The structure-and-bonding foundations behind every chemistry course you will take

FACULTY PRIMER

Faculty Primer

Faculty of Physical Sciences
Pure Chemistry | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Independent study support | Not an official UNIZIK publication

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Why this pack exists

CHM 101 rewards students who can move between the microscopic picture and the calculation. This pack connects atomic structure, periodicity, bonding and stoichiometry, then shows how to set up mole calculations that do not collapse under exam pressure.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Describe atomic structure, electronic configuration and periodic trends with correct reasoning.
- Predict bond type and molecular shape using electronegativity and VSEPR theory.
- Balance equations and solve mole, limiting-reagent and concentration problems.
- Explain the behaviour of acids, bases and buffers at an introductory level.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

Atomic structure and periodicity

The periodic table is not a chart to memorise; it is a map of electron configuration.

- Atomic number is the number of protons; mass number is protons plus neutrons.
- Isotopes have the same atomic number but different mass numbers.
- Electrons fill orbitals in order of increasing energy, following the Aufbau principle, Pauli exclusion principle and Hund's rule.
- Atomic radius decreases across a period and increases down a group.
- Ionisation energy increases across a period and decreases down a group.

- Electron affinity and electronegativity follow similar trends to ionisation energy.

> **Explain, do not just state**
> A question asking why ionisation energy increases across a period wants the effective nuclear charge argument, not a restatement of the trend.

Bonding and molecular shape

Bonding is about electrons: transfer creates ions, sharing creates covalent bonds, and unequal sharing creates polarity.

- Ionic bonding: metal to non-metal, lattice structures, high melting points.
- Covalent bonding: shared electron pairs, discrete molecules or giant networks.
- Metallic bonding: delocalised electrons explain conductivity and malleability.
- Intermolecular forces: London dispersion, dipole-dipole and hydrogen bonding, in increasing strength.
- VSEPR: electron pairs repel and adopt the arrangement that maximises separation.
- Polar bonds do not guarantee a polar molecule; symmetry can cancel the dipoles.

Moles and stoichiometry

Almost every calculation in CHM 101 starts with converting something into moles.

1. Write and balance the equation first.
2. Convert given mass, volume or concentration into moles.
3. Use the balanced equation ratio to find the moles of the required substance.
4. Identify the limiting reagent when two reactant amounts are given.
5. Convert moles back to the requested unit: mass, volume of gas at stated conditions, or concentration.
6. Check significant figures and units.

> **The limiting reagent test**
> Divide the moles of each reactant by its coefficient. The smallest quotient identifies the limiting reagent - not the smallest mass.

Atoms, moles and stoichiometry

Chemistry is quantitative. The mole is the bridge between the microscopic and the measurable: one mole contains Avogadro's number of particles, and its mass in grams equals the relative molecular mass. Almost every calculation in first-year chemistry runs through the mole.

Stoichiometry uses the balanced equation as a ratio machine. Balance first, convert masses to moles, apply the ratio from the coefficients, convert back to the requested quantity. The limiting reagent is the one that runs out first, and it determines the maximum yield; percentage yield compares what you got with that maximum.